

**Panorama Stitching Using SIFT and
ORB Feature Matching with
Unsupervised Machine Learning
Comparison**

(Project Report for Computer Vision A)

BY

THU TA ZAW

1.INTRODUCTION

Overlapping multiple images to combine into a single seamless one by detecting salient visual features, matching them between different images, estimating geometric transformations, and finally warping multiple images into a common coordinate frame is called image stitching. This can be used in various computer vision applications such as smartphones photography, autonomous driving, and robotics. In this project, image stitching method is used to create complete panorama pictures based on both SIFT (Scale-Invariant Feature Transform) and ORB (Oriented FAST and Rotated BRIEF) feature detectors and descriptors.

2.OBJECTIVES

The primary objectives of this project are:

- To implement the complete panoramas picture using the SIFT and ORB feature detectors and descriptors.
- To compare the performance of SIFT and ORB by using the unsupervised machine learning (K-means clustering).
- To analyze the robustness, accuracy, and limitations of SIFT and ORB in real image stitching.

3.METHODOLOGY

The project is written in python programming language. The OpenCV library is used for the image processing. The process starts with choosing the images with identical features and strong overlapping regions. Both SIFT and ORB approach follow the same processing steps: convert images to grayscale, detect key points, compute descriptors, perform descriptor matching using kNN + Lowe's ratio test, compute homography using RANSAC, Warp image 2 into the coordinate frame of image 1 compose the final panorama.

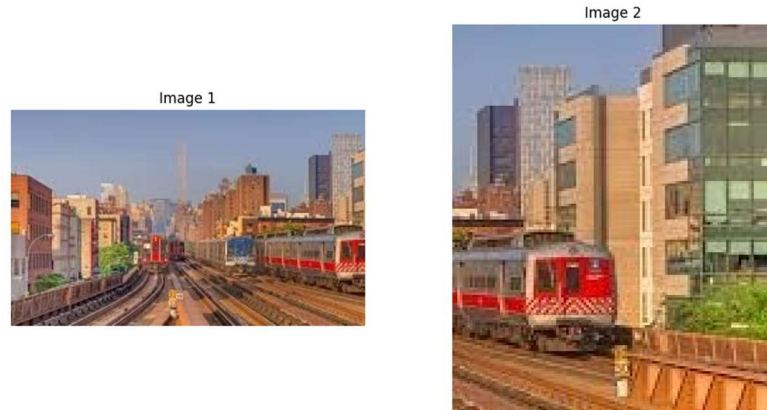


Fig.1 Original images

3.1 SIFT-Based Stitching

The SIFT is implemented based on the classical pipeline introduced by David Lowe and the goal is to extract scale-invariant and rotation-invariant features that provide stable correspondences between the two overlapping images. This begins by detecting the key points using the OpenCV SIFT library and extracting their corresponding 128-dimensional gradient descriptors. FLANN KD-tree matcher is used to perform the descriptor and Lowe's ratio test is used to remove ambiguous matches. The RANSAC method is used to estimate a homography from the remaining matches, and the second image to be warped into the coordinate frame of the first.



Fig.2 SIFT Key points from images

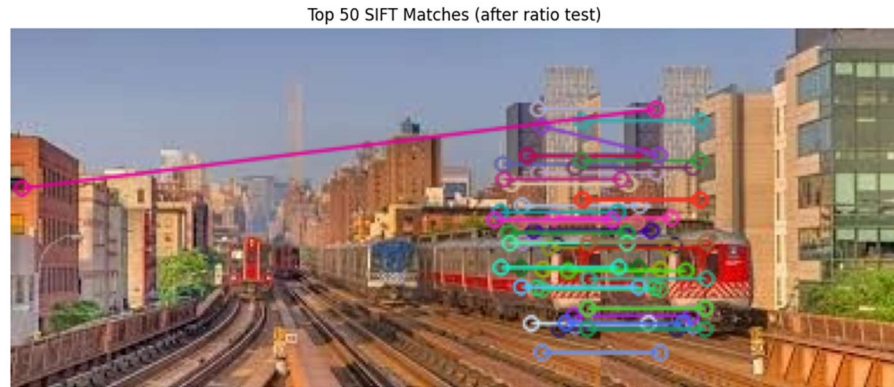


Fig.3 SIFT Key points matching



Fig.4 Final Panorama image from SIFT and Homography

3.2 ORB-Based Stitching

The ORB method is implemented by using OpenCV ORB library and computes 32-byte binary BRIEF descriptors. These descriptors are matched using a brute-force Hamming-distance matcher, with a ratio test applied to refine correspondences. The same RANSAC method is used to compute the homography from the filtered matches, followed by warping and stitching to generate the panorama. Metrics computed for ML Comparison

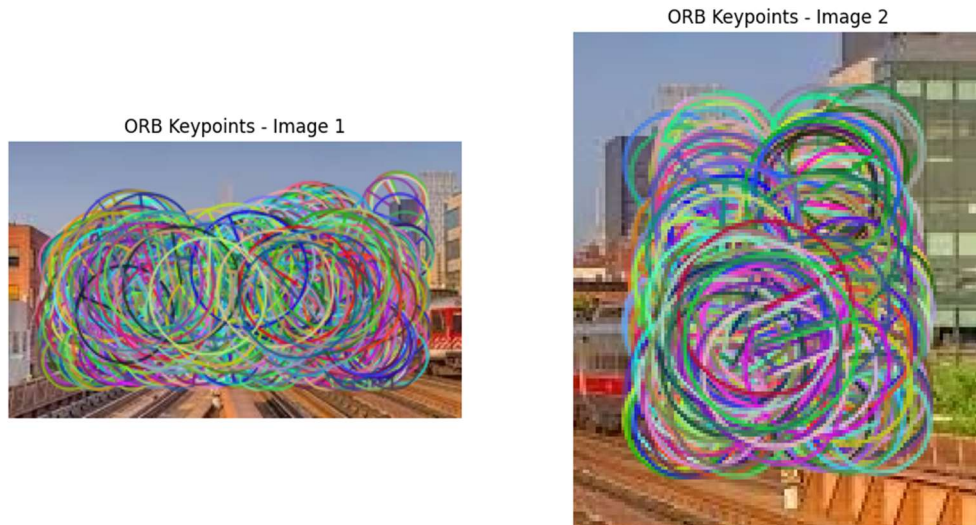


Fig.5 ORB key points from images

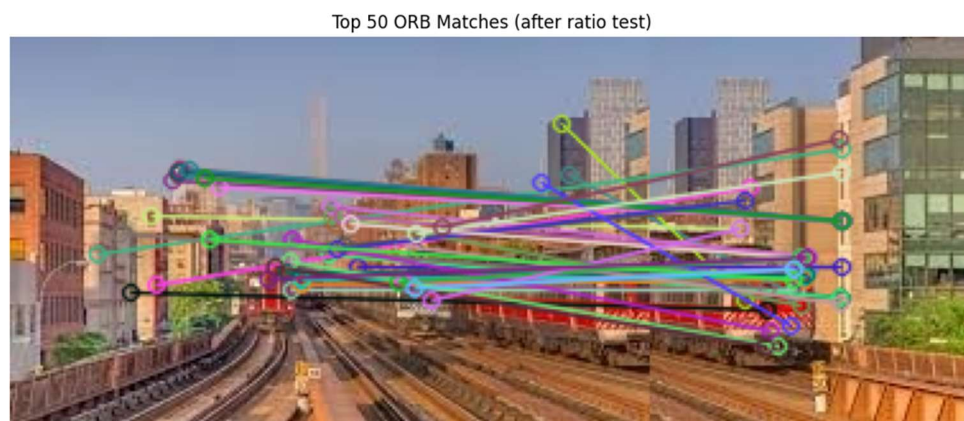


Fig.6 ORB key points matching

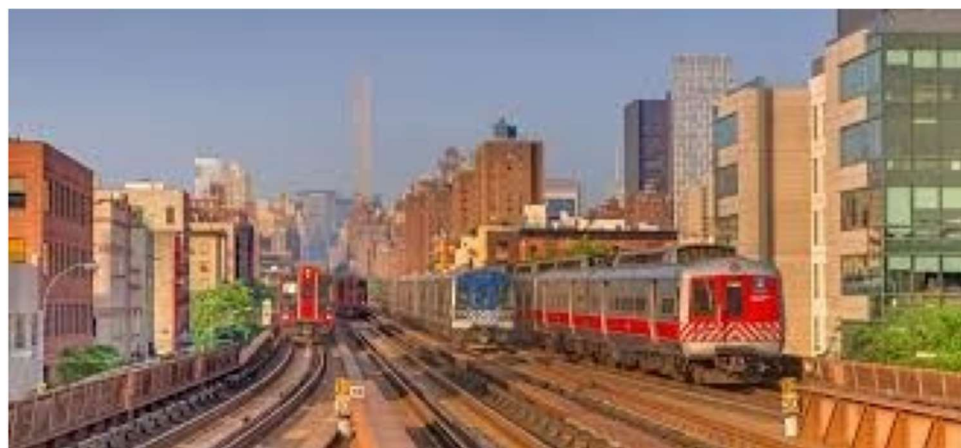


Fig.7 Final Panorama image from ORB and Homography

3.3 Unsupervised ML Evaluation

Several quantitative metrics are extracted from each method to compare the performance of SIFT and ORB methods. Those include the number of good matches, RANSAC inlier ratio, mean descriptor distance, and the mean projection error, which are then combined into feature vectors and normalized before applying K-means clustering. The cluster with high overall performance characteristics was interpreted as the better-performing method. Based on their matching and geometric alignment quality, this unsupervised learning approach provides an objective comparison between SIFT and ORB methods.

4. Results and Discussion

The SIFT implementation produced a high number of key points and reliable feature matches, resulting in a well-aligned and visually consistent panorama, while ORB detected fewer key points overall and generated fewer stable matches leading to a stitched output with small alignment inconsistencies near the overlapping regions. SIFT showed a higher inlier ratio and lower reprojection error, indicating more accurate geometric alignment compared to ORB.

The results showed that the SIFT method produced a larger number of strong matches that survived the ratio test and RANSAC filtering, which contributed to more accurate homography estimation. Even though ORB was fast, it was more sensitive to texture and scale variations. The unsupervised machine learning algorithm comparison validated these results by assigning SIFT to the higher-performing cluster.

5. Conclusion

In conclusion, the ORB remains a faster and lightweight alternative but shows limitations in accuracy under certain conditions, while SIFT provides more stable matches and superior stitching quality for the tested images. The integration of K-means clustering provided an objective, data-driven way to estimate the two methods, confirming SIFT's advantages. This project successfully compared the two classical feature-based methods by using simple machine learning techniques.

THU TA ZAW
2415121095